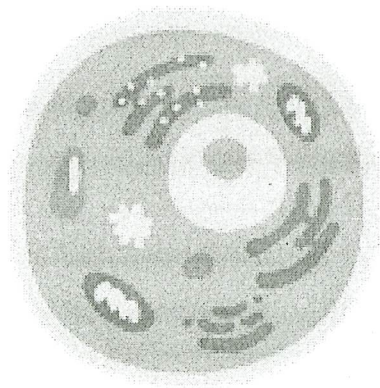
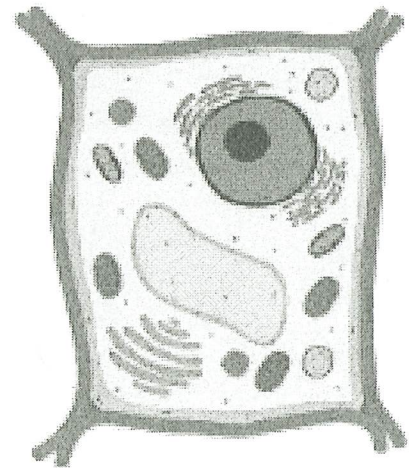
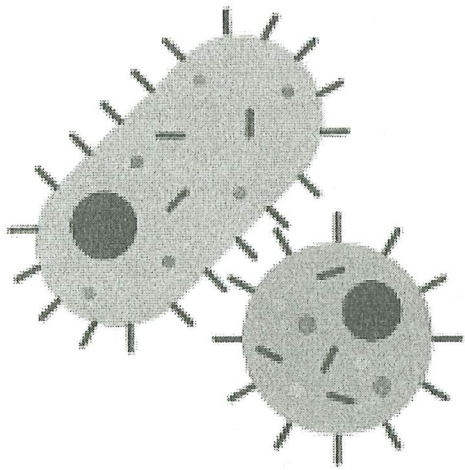


Unit 3 Part 2:

Cell Theory and Structure



Topics:

History of the Cell

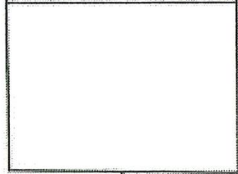
Cell Structure

Organelle Systems

WHAT IS THE

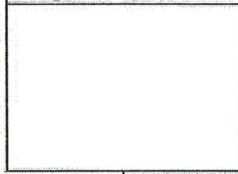
cell theory?

ROBERT HOOKE



1665

ANTON VAN LEEUWENHOEK



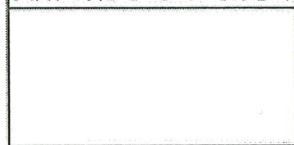
1673

1838

1839

1855

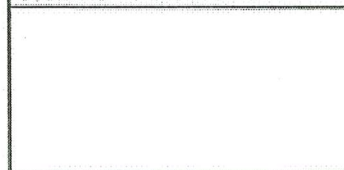
MATTHIAS SCHLEIDEN



RUDOLF VIRCHOW



THEODOR SCHWANN



WHAT ARE THE COMPONENTS OF THE CELL THEORY?

1.

2.

3

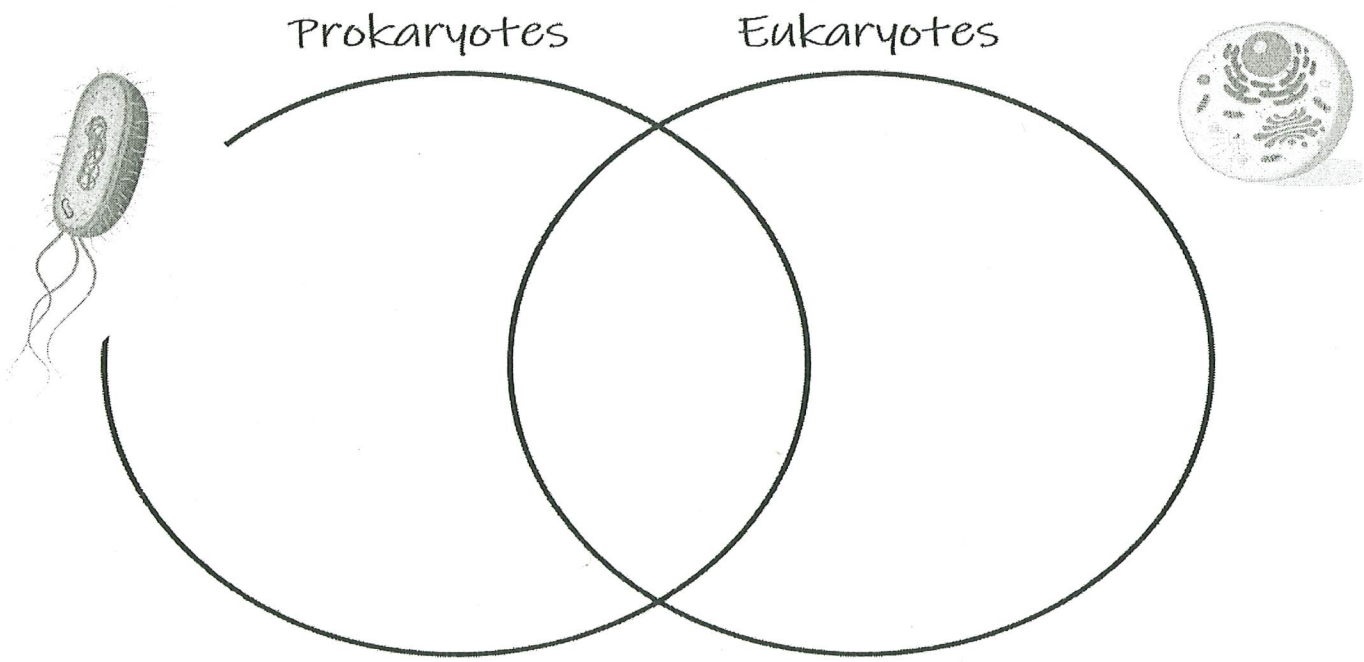


?? HOW DO YOU THINK THE INVENTION OF THE MICROSCOPE INFLUENCED THE CELL THEORY? ??

What do ALL cells have?

	Prokaryotes	Eukaryotes
Type of Organism		
Size		
Nucleus?		
DNA?		
Cell Wall?		
Membrane bound organelles?		
Ribosome Size		
Method of Reproduction?		

Compare Eukaryotes and Prokaryotes below using the Venn Diagram



The 3 Jobs of Cells

1.	2.	3.
----	----	----

Eukaryotic Cell Comparisons

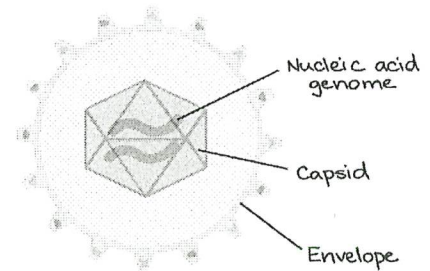
	Plant Cells	Animal Cells
Cell Wall		
Shape		
Vacuoles	Structure: Function:	Structure: Function:
Specialized Organelle		

Animal Cell Surfaces

<u>Animal Cell Surfaces</u>	<u>Plant Cell Surfaces</u>
-----------------------------	----------------------------

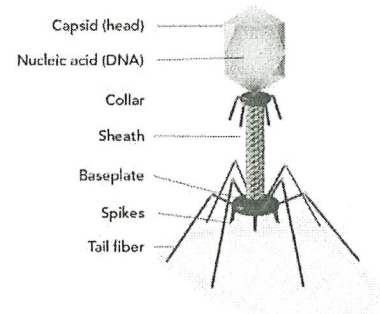
Viruses

- **Description:**
- **Size:**
- **Are they living?:**



- **Features of viruses:**

Structure of a bacteriophage



- **Reproduction:**

Practice EOC Questions

- How is a prokaryotic bacterium different from a eukaryote?
 - It has ribosomes to make proteins
 - It stores genetic information in DNA
 - It has no membrane bound nucleus
 - It has a cell membrane
- Which of these statements best summarize cell theory?
 - Cell contain a nucleus and organelles
 - Cells come in different shapes and sizes
 - Cells can be seen through a microscope
 - Cells are the building blocks of life
- When Robert Hooke observed cork through his microscope, all he could see were empty chambers. Why were these chambers empty?
 - His microscope was not powerful enough to see into the cells
 - Cork comes from plants and plants do not have organelles
 - He was observing structures within the cell and not the whole cell
 - Cork is made of dead plant cells, so only the cell wall remained
- Which of these statements is a principle of cell theory that supports the idea that a cut will heal?
 - All living things are made of cells
 - Cells are the basic unit of structure and function in living things
 - All presently existing cells arise from previously existing cells
 - Most cells are too small to be viewed without a microscope
- Which of the following organisms would be considered an exemption to cell theory?
 - Bacteria
 - Mushroom
 - Virus
 - Amoeba

ORGANELLE SCAVENGER HUNT

Directions: Search around the room to find all organelles! Be DETAILED with your descriptions. In the boxes circle which organisms the organelle is found in.

Organelle	Structure	Function
Cell (Plasma) Membrane Plant Animal Bacteria		
Cell Wall Plant Animal Bacteria	In Plants: In Fungi: In Bacteria:	
Cytoskeleton Plant Animal Bacteria		Microfilaments: Microtubules: Intermediate Filaments:
Cytosol Plant Animal Bacteria	Cytoplasm:	
Nucleus Plant Animal Bacteria		
Nucleolus Plant Animal Bacteria		
Ribosomes Plant Animal Bacteria		Free: Bound:
Rough Endoplasmic Reticulum (ER) Plant Animal Bacteria		

Smooth Endoplasmic Reticulum (ER) Plant Animal Bacteria		
Golgi Apparatus Plant Animal Bacteria		
Vacuoles (Animals) Plant Animal Bacteria		Contractile Vacuole:
Central Vacuole Plant Animal Bacteria		
Vesicles Plant Animal Bacteria		
Lysosomes Plant Animal Bacteria		
Mitochondria Plant Animal Bacteria		
Chloroplasts Plant Animal Bacteria		
Centrioles Plant Animal Bacteria	Centrosome:	
Cilia Plant Animal Bacteria		
Flagella Plant Animal Bacteria		

Endomembrane System:

What is the endomembrane system?

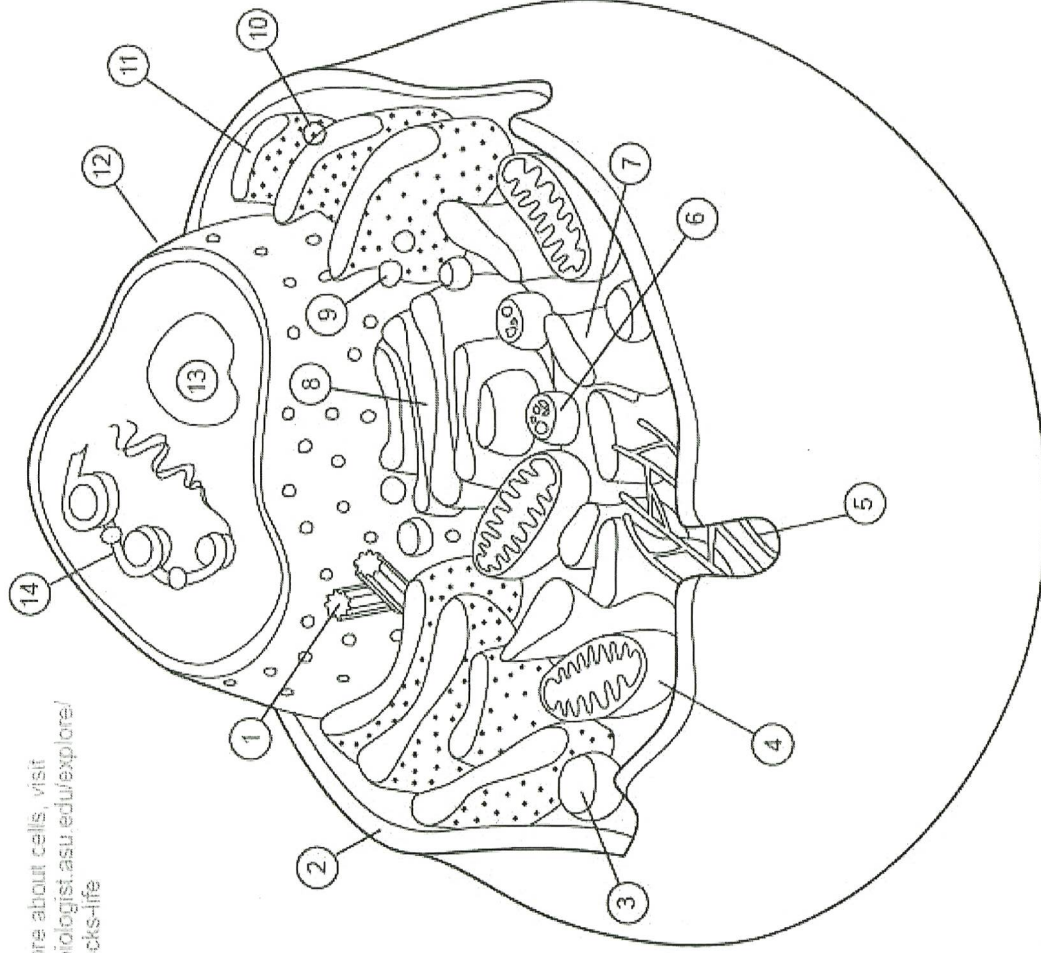
What do they all have in common?

Practice Questions

1. What type of macromolecule is primarily found in cell walls?
2. Which organelle do you think would work closely with the rough ER? Why?
3. What organelle helps construct phospholipids needed by the cell membrane?
4. Which organelle(s) would generate many vesicles and why?
5. Why do plant cells have chloroplast but animal cells do not?
6. Which of the organelles above are associated with generating proteins?
7. Why is a plant's vacuole much larger than an animal's smaller vacuoles?

Cell Labeling Practice

To learn more about cells, visit
<http://askabiologist.asu.edu/explore/building-blocks-life>



Color and label parts:

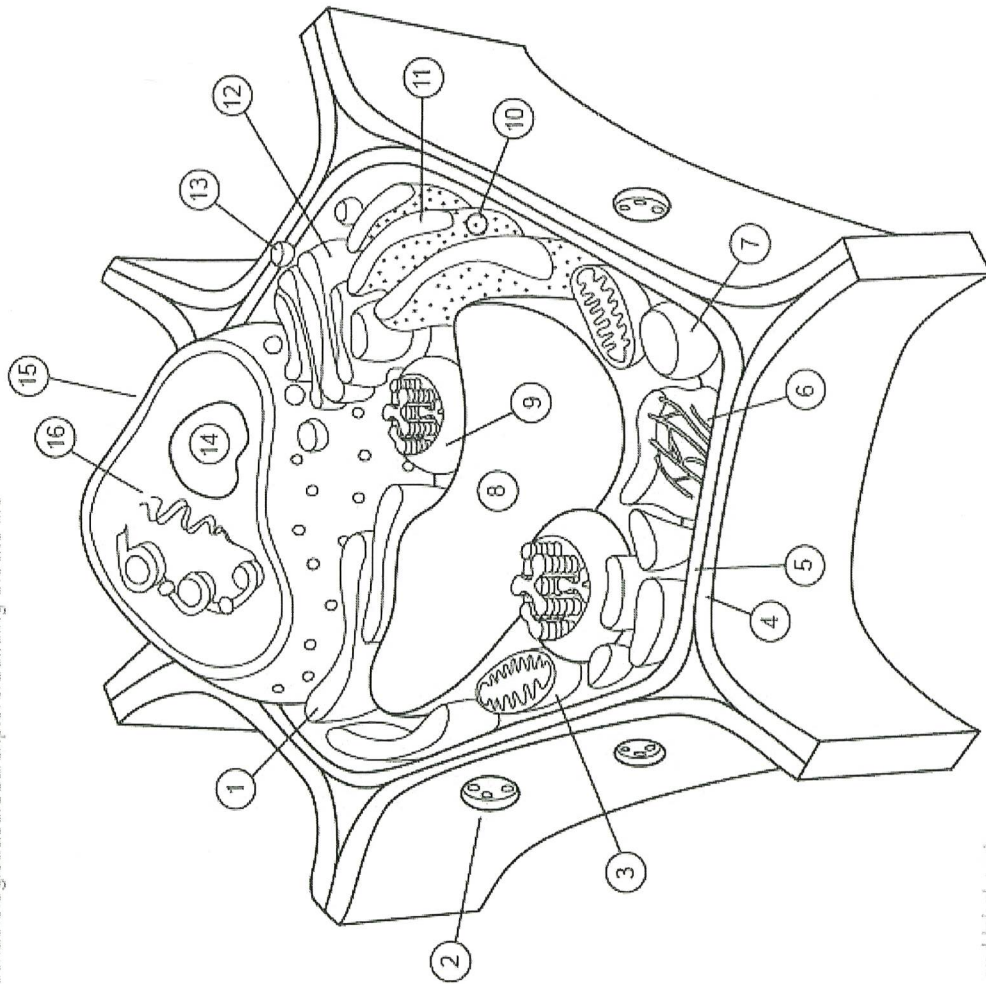
1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____

Word Bank (for ALL cells)

- Vacuole
- Central Vacuole
- Lysosome
- Nucleus
- Cilia
- Flagellum
- Rough ER
- Smooth ER
- Mitochondria
- Chloroplast
- Nucleolus
- Centrioles
- Plasmodesmata
- Vesicles
- Nucleoid Region
- Cytoskeleton
- Cell Wall
- Capsule
- Plasma Membrane
- DNA
- Ribosomes
- Cytosol
- Plasmid

Plant and animal cells have many common parts, but a few are unique to plant cells. Do you know which parts are different? To learn more about cells, visit <http://askabiologist.asu.edu/explore/building-blocks-life>



Color and label parts:

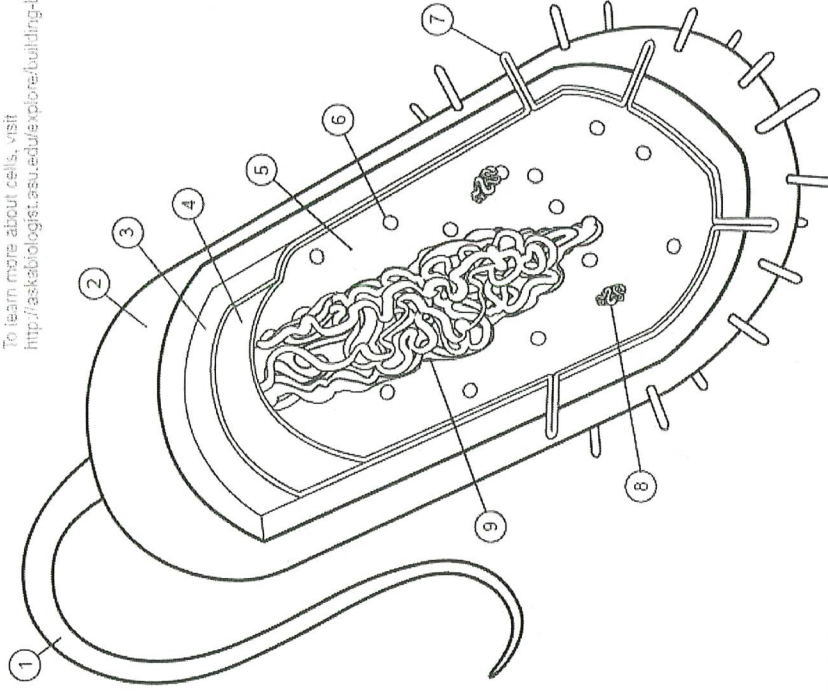
1. _____
2. _____
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6. _____
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8. _____

9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____

Bacteria are the oldest life forms on Earth that we know of, and they outnumber all other organisms on Earth.

To learn more about cells, visit

<http://askabiologist.asu.edu/explore/building-blocks-life>



Color and label parts:

1. _____
2. _____
3. _____
4. _____
5. _____

6. _____
7. _____
8. _____
9. _____

Organelle Hexagonal Thinking Activity

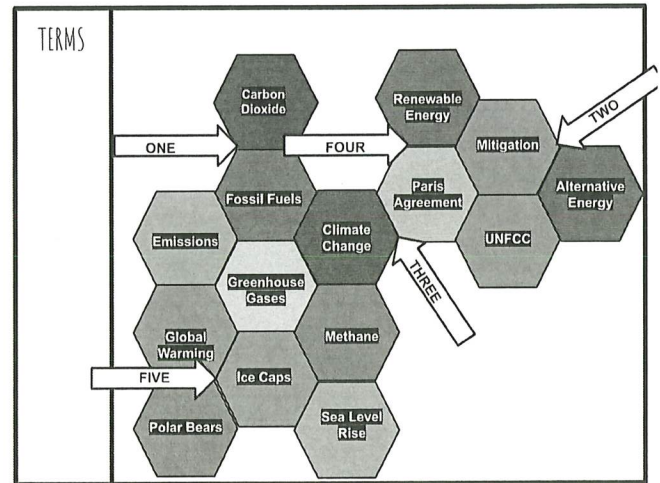
Using your list of organelles you were given in class and a set of hexagons, you will be creating a matrix to represent how different organelles work together in order to maintain homeostasis in the cell. You will need to be able to explain WHY you connected specific organelles together.

Step 1:

Use your organelle information sheet from the scavenger hunt and write names of organelles on the hexagons. *Note you may not end up using all of the organelles.

Step 2: Put them together in a matrix to make connections (see example at the right- yours may look different).

Step 3: Use arrows to point out 5 connections you made. Record this information in the table below. Make sure to use your knowledge of organelles to thoroughly explain WHY these specific organelles can be connected (they make a product, work together, etc). DO NOT REPEAT ORGANELLES- find 5 different connections.



Organelles	Why did you make this connection? EXPLAIN
Organelle 1: Organelle 2:	
Organelle 1: Organelle 2:	
Organelle 1: Organelle 2:	
Organelle 1: Organelle 2:	

Organelle 1: Organelle 2:	
------------------------------	--

Hexagonal Activity Analysis Questions:

1. What organelles are responsible for producing proteins (NAME ALL)? What is the name for this organelle system?
2. Why is the smooth endoplasmic reticulum important to the cell? What does it produce and how are its products used in the cell?
3. What would be a consequence of a plant having a dysfunctional chloroplast?
4. A cell is observed under a microscope and contains a large number of mitochondria. What might this suggest about the function of the cell, and why? What would an example of a cell with many mitochondria be?
5. Prokaryotic cells lack membrane-bound organelles, while eukaryotic cells contain them. How do membrane-bound organelles give eukaryotic cells an advantage in terms of efficiency and complexity?

Student Checklist

Names: _____ Period: _____

1. Accuracy of Connections

- ☐ Did I make 5 connections between organelles?
- ☐ Are my connections scientifically correct?
- ☐ Do my organelles actually work together or depend on each other in the way I described?

2. Explanation of Connections

- ☐ Did I explain how the two organelles are connected?
- ☐ Did I give enough detail so someone else could understand the relationship?
- ☐ Did I avoid one-word or vague answers?

3. Complexity & Thoughtfulness

- ☐ Did I go beyond simple facts (ex: not just "they are both in the cell")?
- ☐ Did I explain the **process** or the reason they depend on each other (ex. protein synthesis, energy transfer, or waste removal)?
- ☐ Did I show deeper thinking about how the cell works as a system?

4. Completion & Analysis Questions

- ☐ Did I answer all of the analysis questions at the end?
- ☐ Did I check that my answers are accurate?
- ☐ Did I show reflection (not just short/simple answers)?

Teacher Rubric

	5 Points	4 Points	3 Points	2 Points	0-1 Points	Total
Accuracy of Connections	All connections are scientifically accurate and clearly show how the organelles interact or depend on one another.	Most connections are accurate, with only one minor error or unclear link.	Some connections are accurate, but others are vague, incomplete, or incorrect.	Only 1–2 connections show accurate relationships.	Very few or no connections are accurate.	___/5
Explanation of Connections	Each connection is thoroughly explained with clear reasoning, showing strong understanding of cell processes.	Most explanations are clear and logical, but with a slight lack of detail.	Some explanations are clear, but others are too brief or lack reasoning.	Explanations are minimal, unclear, or repetitive.	Explanations are missing or do not make sense	___/5
Complexity and Critical Thinking	Connections show higher-level thinking, going beyond surface-level facts to explain how organelles work together in systems (e.g., protein synthesis, energy flow, waste removal).	Connections mostly show good thought, with some deeper connections but also some surface-level ones.	Connections show basic understanding, but few demonstrate deeper thinking.	Connections are mostly simple or obvious, with little attempt at depth.	Very little thought or originality shown.	___/5
Analysis Questions	All analysis questions are complete, accurate, and demonstrate reflection/understanding.	Most questions are complete and accurate, with some minor gaps.	Some questions are incomplete, vague, or have errors.	Only a few questions are answered, with limited accuracy or effort.	Questions not completed.	___/5
						___/20

Microscope use:

1. Place the Slide on the Microscope _____
2. Use Stage _____ or _____
3. Always start on the _____, it is there for referencing specific areas on the slide.
4. Look into the _____, you should see a _____
5. Use the _____ first, this will be the larger knob
6. Use the fine focus _____ to make the image clear. This will be the smaller knob. You should be able to see a cell or a recognizable object.
7. Click the objective to the next highest power, get the image in _____, you should here the objective
8. When you get to the largest objective, do _____ use the coarse focusing knob, use the fine focus knob ONLY

Microscope Lab

You will be viewing various prepared slides under the microscope.

For each slide you will need to

- start with the lowest (smallest) objective and get the object into focus
- move to the next highest objective and get the object into focus
- write down what you are viewing using the label on the slide
- draw what you can see, your drawing should not be dots
- Label what you see

Draw and label what you see, make sure you label the magnification under each picture:

Microscope slide label _____

Draw and label what you see, make sure you label the magnification under each picture:
Microscope slide label _____

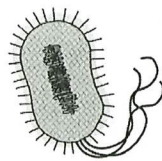
Draw and label what you see, make sure you label the magnification under each picture:
Microscope slide label _____

Draw and label what you see, make sure you label the magnification under each picture:
Microscope slide label _____

Questions:

1. Why should you always begin with the lowest magnification?
2. What is the general shape of a typical plant cell? Animal cell?
3. Why are the prepared slides you viewed stained a certain color?
4. Why must the specimen you are viewing be only 1 cell layer thick?
5. What was the most interesting thing you saw today?

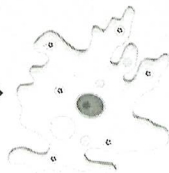
Bacteria (*E. coli*)



Visible Features:

- Cell wall (if stained)
- Flagella (if stained)
- ~1 – 10 μm

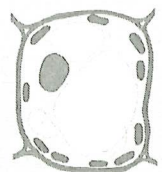
Protist (*Amoeba*)



Visible Features:

- Nucleus
- Pseudopodia
- Food vacuoles
- ~50 – 500 μm

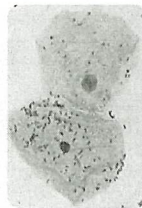
Plant Cell (Leaf)



Visible Features:

- Nucleus
- Chloroplasts
- Cell wall
- ~10 – 100 μm

Animal Cell (cheek)



Visible Features:

- Nucleus
- Mitochondria (only if stained)
- ~10 – 50 μm

Cell Investigation Stations

Directions: As you move through each of the stations, answer the questions and take notes as needed.

Station 1: Macromolecules and Organelles

Questions:

1. Mitochondria function to provide us with energy in the form of ATP from by breaking down glucose. Which type of macromolecule would you find in the most quantity in mitochondria and WHY?

Macromolecule:

Reasoning:

2. Ribosomes are made of two subunits of rRNA that get assembled in the nucleolus. What type of macromolecule would be found most in ribosomes and WHY?

Macromolecule:

Reasoning:

3. The cell membrane is made up of long chains of fatty acids that help regulate what moves in and out of the cell. What type of macromolecule would you find most in the cell membrane and WHY?

Macromolecule:

Reasoning:

4. The cytoskeleton of the cell has diverse functions. Not only is it helpful in stabilizing the cell, but it also can help provide movement in some single-celled Eukaryotes. It consists of a network of filaments that are composed of polypeptides. What macromolecule would you find most in the cytoskeleton and WHY?

Macromolecule:

Reasoning:

5. In what organelle would we find almost nothing but nucleic acids? Why?

Organelle:

Reasoning:

6. Which of our organelles is specialized in producing lipids for the cell membrane?

Organelle:

Station 2: Human Cells

Red Blood Cells:

1. What is the function of red blood cells?
2. How does the structure (shape of the red blood cell) relate to its function?

3. What is the function of a lymphocyte?
4. Why do you think the lymphocyte is larger and rounder than the red blood cell?
5. List the organelles missing in red blood cells.
6. Most cells fix broken organelles, but in red blood cells, they are destroyed after a certain amount of damage. Explain why red blood cells cannot replace damaged proteins.

Nerve Cells:

1. What is the job of nerve cells?
2. How does the nerve cell structure relate to its function?
3. Nerve cells are long and covered in a fatty covering that helps to conduct signals quickly from nerve cell to nerve cell.
 - a. What organelle is responsible for producing this fatty covering?
 - b. What would happen to the function of nerve cells if this organelle was damaged or absent?

Sperm Cells:

1. Look at the diagram of the sperm cell. What are its 3 largest organelles and what are their jobs?
2. How do these organelles relate to the function of the sperm cell and why are they important?

Station 3: Organelles and Disorders

Cystic Fibrosis (CF)

- a. Which organelle or structure might not be able to perform its job as a result of this genetic disorder?
- b. How did you determine which organelle it was?

Muscular Dystrophy

- a. Which organelle or structure might not be able to perform its job as a result of this genetic disorder?
- b. How did you determine which organelle it was?

I-Cell Disease

- a. Which organelle or structure might not be able to perform its job as a result of this genetic disorder?

- b. How did you determine which organelle it was?

Tay Sach's Disease

- a. Which organelle or structure might not be able to perform its job as a result of this genetic disorder?
- b. How did you determine which organelle it was?

Reflection:

Why can mutations in our DNA be harmful to how our cells function?

Station 4: Identifying Types of Transport

- A. Type of transport: _____ Active or Passive? _____
- B. Type of transport: _____ Active or Passive? _____
- C. Type of transport: _____ Active or Passive? _____
- D. Type of transport: _____ Active or Passive? _____
- E. Type of transport: _____ Active or Passive? _____
- F. Type of transport: _____ Active or Passive? _____

Station 5: Oddballs of the Natural World- Protists

Paramecium

1. What structures or behaviors does this protist have?
2. Based on its description, is this an example of a plant or animal cell? Explain your reasoning.

Euglena

1. What structures or behaviors does this protist have?
2. Based on its description, is this an example of a plant or animal cell? Explain your reasoning.

Noctiluca

1. What structures or behaviors does this protist have?
2. Based on its description, is this an example of a plant or animal cell? Explain your reasoning.

Reflection

Many of the protists above defy classification into only one category, challenging our conceptions of how we classify organisms. How would you break down protists in order to classify them?

Station 6- Organelles that Build Proteins

Summarize the steps to making a protein:

Step 1: Proteins are made by _____

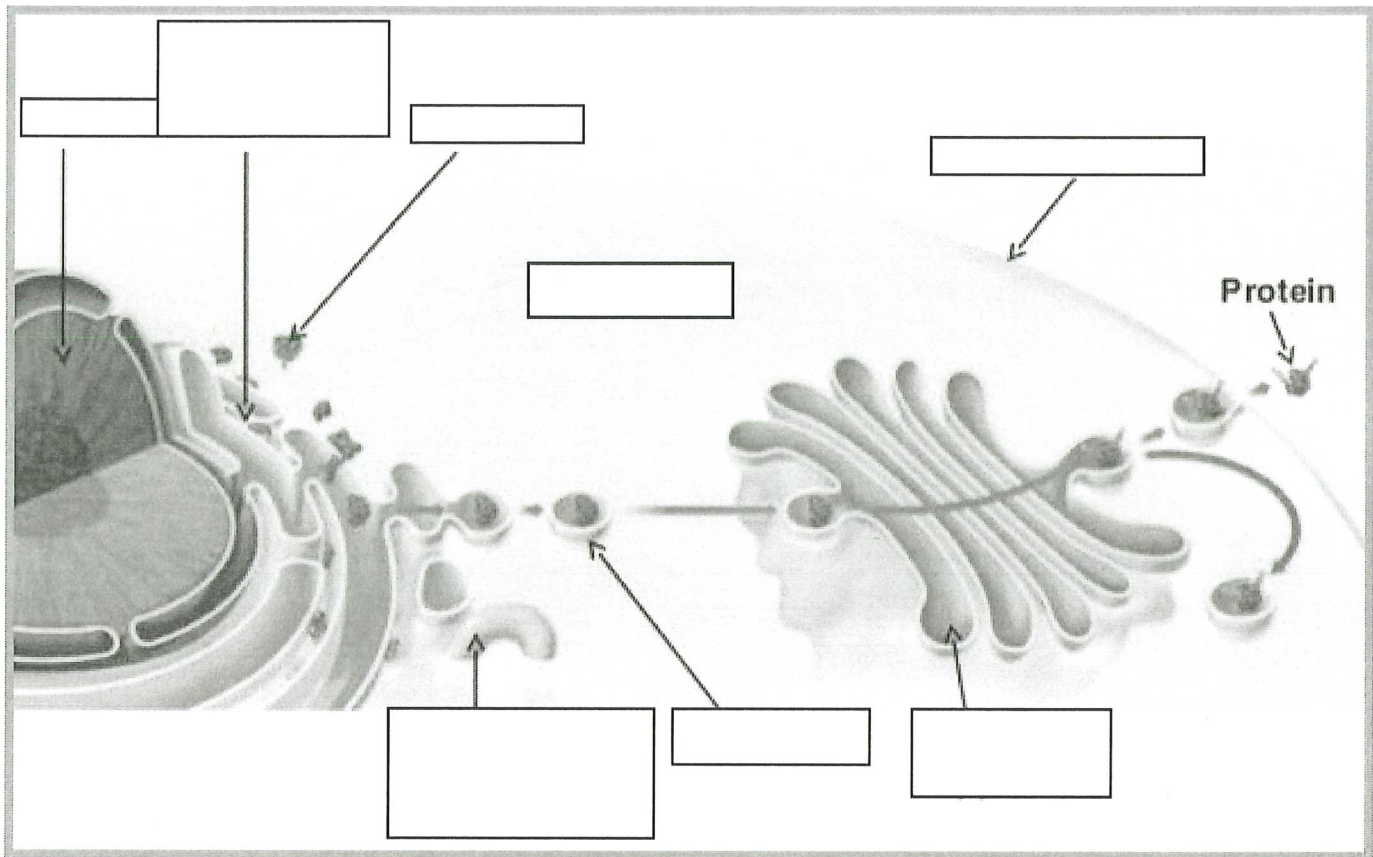
Step 2: Protein folded on the _____

Step 3: Transported in a _____ to the _____

Step 4: Processed in the _____ for finishing touches and packaging

Step 5: Transported by another _____ to the _____ to be exported

Label the diagram below that summarizes protein synthesis:



Questions:

1. What would happen if the ribosome stopped functioning correctly?
2. If the cell were missing vesicles, how would this affect the process of making a protein?
3. Where are the instructions to making a protein located inside the cell?
4. What do you think would happen to the protein if there were errors in the cell's DNA?

Station 7- Identifying Cells

- | | |
|--------------------------------|--------------------------------|
| 1. Type of cell:
Reasoning: | 5. Type of cell:
Reasoning: |
| 2. Type of cell:
Reasoning: | 6. Type of cell:
Reasoning: |
| 3. Type of cell:
Reasoning: | 7. Type of cell:
Reasoning: |
| 4. Type of cell:
Reasoning: | |